Yifan Kang

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Education

Massachusetts Institute of Technology *(B.S., Mathematics and Computer Science)* *2024 – Present*

- GPA: 5.0/5.0 **Selected Coursework:** Deep Learning (6.7960), Machine Learning (6.7900), NLP (6.4610), Robotics (6.4200), Robotics Manipulation (6.4210), Advanced Algorithms (6.5210).

Research Experience

Research Intern @ 2077AI *Jun. 2026 – Present*

- Built **ENCORE**: a coding agent generates manipulation programs from **3 demonstrations** and self-repairs from **execution traces**, achieving **95.1%** success on LIBERO-PRO.
- Fine-tuned **vision-language-action (VLA) models with LoRA** for real-robot manipulation and validated the resulting policies on physical hardware.
- Deployed **robot-use agents on ALOHA** for long-horizon bimanual manipulation and evaluated task execution on real robots.

Student Researcher @ MIT *(Prof. Leslie Pack Kaelbling)* *Jun. 2025 – Present*

- **KeyFlow (CoRL 2026 Workshop Learn@Deploy):** Built a coding agent that adapts a manipulation policy by revising how a handful of demonstrations are represented and reused: **VLM**-grounded 3-D semantic keypoints plus short programs for the state, the inter-state distance, and motion transfer, refined from simulated play with no new demonstrations or weight updates.
- **62.6%** mean held-out success on 13 RoboCasa tasks vs. 47.8% for a $\pi_{0.5}$ VLA fine-tuned on 10× more demonstrations; composing its flow fields with a base $\pi_{0.5}$ lifts mean success by **+37.2 points** across 6 IsaacSim tasks, improving every one.

Student Researcher @ MIT *(Prof. Antonio Torralba)* *Jun. – Aug. 2025*

- Co-first author of *Doodle Agent: Exploring Freeform Visual Generation with Multimodal LLMs* (Accepted to **ICCV 2025 AI4VA Workshop**).

Interpreting Diffusion Language Models Over Denoising Time *2026*

Trained sparse autoencoders and cross-layer transcoders on diffusion LLMs, with timestep-aware tracing and causal interventions (Accepted to **CoLM 2026 Workshop on Non-Autoregressive LMs**).

Selected Publications

- *KeyFlow: Agentic Policy Adaptation with Episodic Memory.* **CoRL 2026 Workshop Learn@Deploy.**
  B. Zandonati*, **Y. Kang***, T. Lozano-Pérez, L. P. Kaelbling. (*Co-first authors)
- *ENCORE: Few-Shot Agentic Discovery of Manipulation Strategies.* **CoRL 2026 Workshop on Agentic Robotics; NeurIPS 2026 Robot Learning Workshop (Oral, 6/43).**
  Y. Kang, Z. Wang, Z. Fan, B. Liu.
- *Generalized Planning Interfaces for Agentic Robotics.* **CoRL 2026 Workshop on Agentic Robotics.**
  B. Zandonati, **Y. Kang**, T. Lozano-Pérez, L. P. Kaelbling.

Selected Projects

Autonomous Racecar *(MIT Robotics: Science & Systems)* *Feb. – May 2026*

- Built a full autonomy stack on a **ROS2 racecar** (LiDAR, camera, odometry): particle-filter localization, path planning, and pure-pursuit control; final challenge combined high-speed lane keeping with **YOLO**-based autonomous parking and traffic-light handling.

Awards & Honors

William Lowell Putnam Mathematical Competition — Rank 254 2024
United States of America Mathematical Olympiad (USAMO) — Honorable Mention 2023
MIT CSAIL Agentic AI Hackathon — Winner (Jane Street Award) 2026
United States of America Computing Olympiad (USACO) — Finalist (Top 26) 2022